

ORF 524 — Midterm Exam

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Notes:

- Duration of examination: 180 minutes.
An additional 15 minutes are given to upload your solutions to Gradescope.
- All parts are equally weighted (5 points).
Please provide as much detail as possible in your answers.
Answers without proper justification will not receive full credit.
- Please start each question on a separate page.
A single PDF file must be submitted to Gradescope.
- Only one page of personal notes are allowed during the examination.
This is otherwise a closed-book and individual examination.

1 Uniform Most Powerful Testing (50 points)

Assume $\mathbf{X} = (X_1, \dots, X_n)^\top \sim \mathbb{P}_\theta$ with $\theta \in \Theta$. Let $L(\mathbf{x}|\theta)$ denote the likelihood function of $\mathbb{P}_\theta$. Recall that a *critical region* (or *rejection region*) $\mathcal{C}$ is a subset of the sample space such that the null hypothesis H_0 is rejected if (and only if) the observed sample $\mathbf{X}$ falls into $\mathcal{C}$.

The celebrated **Neyman-Pearson Lemma** states that for testing $H_0 : \theta = \theta_0$ vs. $H_1 : \theta = \theta_1$ at a significance level $\alpha \in (0, 1)$, the most powerful critical region $\mathcal{C}$ is defined by

$$\mathcal{C}_{\text{NP}} = \left\{ \mathbf{x} : \frac{L(\mathbf{x}|\theta_1)}{L(\mathbf{x}|\theta_0)} > k \right\},$$

where k is chosen such that $\mathbb{P}_{\theta_0}(\mathbf{X} \in \mathcal{C}_{\text{NP}}) = \alpha$. (For the purpose of this question, we focus on the core non-randomized rejection region $\mathcal{C}_{\text{NP}}$ and ignore the boundary case where the ratio equals k , which occurs with $\mathbb{P}_{\theta_0}$ -zero probability when $\mathbf{X}$ is continuously distributed.)

The following four questions will guide in proving the **Neyman-Pearson Lemma**. Let $\mathcal{C}$ be any other critical region such that $\mathbb{P}_{\theta_0}(\mathbf{X} \in \mathcal{C}) = \alpha$.

1. Define the power of a test associated with the critical region $\mathcal{C}$, denoted by $\beta_{\mathcal{C}}(\theta_1)$.

Solution: The power of the test $\mathcal{C}$ at θ_1 is the probability of rejecting H_0 when H_1 is true:

$$\beta_{\mathcal{C}}(\theta_1) = \mathbb{P}_{\theta_1}(\mathbf{X} \in \mathcal{C}) = \int_{\mathcal{C}} L(\mathbf{x}|\theta_1) d\nu(\mathbf{x}).$$

2. Decompose the difference between the rejection regions $\mathcal{C}_{\text{NP}}$ and $\mathcal{C}$ into the disjoint regions $\mathcal{C}_{\text{NP}} \setminus \mathcal{C}$ and $\mathcal{C} \setminus \mathcal{C}_{\text{NP}}$. Use the size constraint to show that: $\beta_{\mathcal{C}_{\text{NP}} \setminus \mathcal{C}}(\theta_0) = \beta_{\mathcal{C} \setminus \mathcal{C}_{\text{NP}}}(\theta_0)$.

Solution: Decompose the rejection regions: $\mathcal{C}_{\text{NP}} = (\mathcal{C}_{\text{NP}} \cap \mathcal{C}) \cup (\mathcal{C}_{\text{NP}} \setminus \mathcal{C})$ and $\mathcal{C} = (\mathcal{C}_{\text{NP}} \cap \mathcal{C}) \cup (\mathcal{C} \setminus \mathcal{C}_{\text{NP}})$. The size constraint implies: $\mathbb{P}_{\theta_0}(\mathbf{X} \in \mathcal{C}) = \mathbb{P}_{\theta_0}(\mathbf{X} \in \mathcal{C}_{\text{NP}})$, and hence

$$\mathbb{P}_{\theta_0}(\mathbf{X} \in \mathcal{C}_{\text{NP}} \cap \mathcal{C}) + \mathbb{P}_{\theta_0}(\mathbf{X} \in \mathcal{C}_{\text{NP}} \setminus \mathcal{C}) = \mathbb{P}_{\theta_0}(\mathbf{X} \in \mathcal{C}_{\text{NP}} \cap \mathcal{C}) + \mathbb{P}_{\theta_0}(\mathbf{X} \in \mathcal{C} \setminus \mathcal{C}_{\text{NP}})$$

Subtracting the common term $\mathbb{P}_{\theta_0}(\mathbf{X} \in \mathcal{C}_{\text{NP}} \cap \mathcal{C})$ yields desired equality.

3. Prove that the integral of $L(\mathbf{x}|\theta_1)$ over the region $\mathcal{C}_{\text{NP}} \setminus \mathcal{C}$ is greater to the integral of $L(\mathbf{x}|\theta_1)$ over the region $\mathcal{C} \setminus \mathcal{C}_{\text{NP}}$. (It would be “greater or equal” if accounting for the boundary case via randomization was needed.)

Solution: Based on the definition of $\mathcal{C}_{\text{NP}}$, the following basic inequalities follow.

- (a) For $\mathbf{x} \in \mathcal{C}_{\text{NP}} \setminus \mathcal{C}$: $\mathbf{x}$ is in $\mathcal{C}_{\text{NP}}$, and hence

$$\frac{L(\mathbf{x}|\theta_1)}{L(\mathbf{x}|\theta_0)} > k \implies L(\mathbf{x}|\theta_1) > kL(\mathbf{x}|\theta_0)$$

(b) For $\mathbf{x} \in \mathcal{C} \setminus \mathcal{C}_{\text{NP}}$: $\mathbf{x}$ is not in $\mathcal{C}_{\text{NP}}$, and hence

$$\frac{L(\mathbf{x}|\theta_1)}{L(\mathbf{x}|\theta_0)} < k \implies L(\mathbf{x}|\theta_1) < kL(\mathbf{x}|\theta_0)$$

Integrate the inequalities to conclude that

$$\begin{aligned} \int_{\mathcal{C}_{\text{NP}} \setminus \mathcal{C}} L(\mathbf{x}|\theta_1) d\nu(\mathbf{x}) &> k \int_{\mathcal{C} \setminus \mathcal{C}_{\text{NP}}} L(\mathbf{x}|\theta_0) d\nu(\mathbf{x}), \text{ and} \\ \int_{\mathcal{C} \setminus \mathcal{C}_{\text{NP}}} L(\mathbf{x}|\theta_1) d\nu(\mathbf{x}) &< k \int_{\mathcal{C} \setminus \mathcal{C}_{\text{NP}}} L(\mathbf{x}|\theta_0) d\nu(\mathbf{x}), \end{aligned}$$

where $\nu(\cdot)$ denotes the appropriate carrying measure. Since the right-hand side integrals involving $L(\mathbf{x}|\theta_0)$ are equal, we conclude that

$$\int_{\mathcal{C}_{\text{NP}} \setminus \mathcal{C}} L(\mathbf{x}|\theta_1) d\nu(\mathbf{x}) > \int_{\mathcal{C} \setminus \mathcal{C}_{\text{NP}}} L(\mathbf{x}|\theta_1) d\nu(\mathbf{x}).$$

Thus, $\int_{\mathcal{C}_{\text{NP}} \setminus \mathcal{C}} L(\mathbf{x}|\theta_1) d\nu(\mathbf{x}) > \int_{\mathcal{C} \setminus \mathcal{C}_{\text{NP}}} L(\mathbf{x}|\theta_1) d\nu(\mathbf{x})$.

4. Prove the **Neyman-Pearson Lemma** by showing that $\beta_{\mathcal{C}_{\text{NP}}}(\theta_1) > \beta_{\mathcal{C}}(\theta_1)$.

Solution: We have

$$\begin{aligned} \beta_{\mathcal{C}_{\text{NP}}}(\theta_1) - \beta_{\mathcal{C}}(\theta_1) &= \left(\int_{\mathcal{C}_{\text{NP}} \cap \mathcal{C}} L(\mathbf{x}|\theta_1) d\nu(\mathbf{x}) + \int_{\mathcal{C}_{\text{NP}} \setminus \mathcal{C}} L(\mathbf{x}|\theta_1) d\nu(\mathbf{x}) \right) \\ &\quad - \left(\int_{\mathcal{C} \cap \mathcal{C}_{\text{NP}}} L(\mathbf{x}|\theta_1) d\nu(\mathbf{x}) + \int_{\mathcal{C} \setminus \mathcal{C}_{\text{NP}}} L(\mathbf{x}|\theta_1) d\nu(\mathbf{x}) \right) \\ &= \int_{\mathcal{C}_{\text{NP}} \setminus \mathcal{C}} L(\mathbf{x}|\theta_1) d\nu(\mathbf{x}) - \int_{\mathcal{C} \setminus \mathcal{C}_{\text{NP}}} L(\mathbf{x}|\theta_1) d\nu(\mathbf{x}) \\ &> 0. \end{aligned}$$

Thus, $\mathcal{C}_{\text{NP}}$ is more powerful than any other size α test $\mathcal{C}$, completing the proof.

For the rest of this question, assume that $\mathbf{X} = (X_1, \dots, X_n)^\top$ is a vector of i.i.d. random variables with marginal distribution $\text{Exponential}(\theta)$, and $\theta_1 > \theta_0$.

5. Write down the joint likelihood function $L(\mathbf{x}|\theta)$ and the likelihood ratio $\Lambda(\mathbf{x}) = \frac{L(\mathbf{x}|\theta_1)}{L(\mathbf{x}|\theta_0)}$.

Solution: The joint likelihood function is:

$$L(\mathbf{x}|\theta) = \prod_{i=1}^n f(x_i|\theta) = \prod_{i=1}^n \left(\frac{1}{\theta} e^{-x_i/\theta} \right) = \left(\frac{1}{\theta} \right)^n e^{-\frac{1}{\theta} \sum_{i=1}^n x_i} = \frac{1}{\theta^n} e^{-T/\theta}$$

where $T = \sum_{i=1}^n X_i$. The likelihood ratio is:

$$\Lambda(\mathbf{x}) = \frac{L(\mathbf{x}|\theta_1)}{L(\mathbf{x}|\theta_0)} = \frac{\frac{1}{\theta_1^n} e^{-T/\theta_1}}{\frac{1}{\theta_0^n} e^{-T/\theta_0}} = \left(\frac{\theta_0}{\theta_1}\right)^n \exp \left[T \left(\frac{1}{\theta_0} - \frac{1}{\theta_1} \right) \right].$$

6. Find the most powerful testing procedure for $H_0 : \theta = \theta_0$ vs. $H_1 : \theta = \theta_1$.

Solution: We will show that the critical region defined by $\Lambda(\mathbf{x}) > k$ is equivalent to a rejection rule based on the sample sum $T = \sum_{i=1}^n X_i$, which will demonstrate that the most powerful test rejects H_0 if $T > c$ for some constant c . By the Neyman-Pearson lemma, the most powerful test rejects H_0 if (and only if)

$$\Lambda(\mathbf{x}) = \left(\frac{\theta_0}{\theta_1}\right)^n \exp \left[T \left(\frac{1}{\theta_0} - \frac{1}{\theta_1} \right) \right] > k.$$

First, take the natural logarithm of both sides:

$$\ln \left[\left(\frac{\theta_0}{\theta_1}\right)^n \right] + T \left(\frac{1}{\theta_0} - \frac{1}{\theta_1} \right) > \ln(k).$$

Second, since $\theta_1 > \theta_0$, the coefficient of T is positive:

$$B = \left(\frac{1}{\theta_0} - \frac{1}{\theta_1} \right) = \frac{\theta_1 - \theta_0}{\theta_0 \theta_1} > 0.$$

Third, isolate T :

$$T > \frac{1}{B} \left[\ln(k) - n \ln \left(\frac{\theta_0}{\theta_1} \right) \right] = c.$$

Since the inequality direction does not change (because $B > 0$), the rejection region is $T > c$. Therefore, the most powerful test rejects H_0 if (and only if) $T = \sum_{i=1}^n X_i > c$.

7. Note that $T = \sum_{i=1}^n X_i \sim \text{Gamma}(n, \theta_0)$. For a test of size α , derive the critical value c in terms of n , θ_0 , and an appropriate quantile of the Gamma distribution.

Solution: The critical value c is determined by the size constraint $\mathbb{P}_{\theta_0}(T > c) = \alpha$. This is equivalent to: $\mathbb{P}_{\theta_0}(T \leq c) = 1 - \alpha$. Let $F_T(t; n, \theta_0)$ be the CDF of the $\text{Gamma}(n, \theta_0)$ distribution. Then, c is the $(1 - \alpha)$ -th quantile: $c = F_T^{-1}(1 - \alpha; n, \theta_0)$.

8. Consider the specific scenario where $n = 1$, $\theta_0 = 1$, and $\alpha = 0.05$. Find the exact numerical value of the critical constant c . Then, calculate the power of this test when the true parameter is $\theta_1 = 2$.

Solution: First, we find the critical value c . For $n = 1$ and $\theta_0 = 1$, $T = X_1$ follows an $\text{Exponential}(1)$ distribution. The size constraint is $\mathbb{P}_{\theta_0=1}(X_1 > c) = 0.05$. The PDF is

$f(x|1) = 1e^{-x}$, and therefore

$$\int_c^\infty e^{-x} dx = [-e^{-x}]_c^\infty = 0 - (-e^{-c}) = 0.05 \implies e^{-c} = 0.05,$$

which gives $c = -\ln(0.05) \approx 2.9957$. The exact critical value is $c = -\ln(0.05)$.

Next, we calculate the power at $\theta_1 = 2$, that is, $\mathbb{P}_{\theta_1=2}(X_1 > c) = \beta(2)$. The PDF is $f(x|2) = \frac{1}{2}e^{-x/2}$, and therefore

$$\beta(2) = \int_c^\infty \frac{1}{2}e^{-x/2} dx = [-e^{-x/2}]_c^\infty = 0 - (-e^{-c/2}) = e^{-c/2}.$$

Substitute $c = -\ln(0.05)$ to conclude that

$$\beta(2) = e^{-(-\ln(0.05))/2} = e^{\ln(0.05)/2} = (0.05)^{1/2} = \sqrt{0.05} \approx 0.2236.$$

9. Find the uniform most powerful (UMP) testing procedure for $H_0 : \theta \leq \theta_0$ vs. $H_1 : \theta > \theta_1$.

Solution: The test structure $T > c$ was derived from the likelihood ratio $\Lambda(\mathbf{x})$ for *any* $\theta_1 > \theta_0$ because the coefficient of T in the exponent, $\left(\frac{1}{\theta_0} - \frac{1}{\theta_1}\right)$, is always positive. This confirms the *Monotone Likelihood Ratio* (MLR) property in the statistic T . The **Karlin-Rubin Theorem** states that when the MLR property holds, the most powerful hypothesis test for $H_0 : \theta = \theta_0$ vs. $H_1 : \theta = \theta_1$ is the same regardless of the specific $\theta_1 > \theta_0$ chosen. Since the test $T > c$ is most powerful for every θ_1 in the composite alternative $H_1 : \theta > \theta_1$, it is *Uniformly Most Powerful* (UMP).

10. Why is the critical value c derived for the simple null $H_0 : \theta = \theta_0$ still the correct critical value to use for the composite null $H_0 : \theta \leq \theta_0$ to guarantee that the UMP test has size α ?

Solution: The size of a test is $\sup_{\theta \in \Theta_0} \beta(\theta)$. First, from the MLR property, the power function $\beta(\theta) = \mathbb{P}_\theta(T > c)$ is a non-decreasing function of θ . Second, it follows that the supremum of the power function over the null space is attained at the boundary point $\theta = \theta_0$. That is,

$$\sup_{\theta \leq \theta_0} \beta(\theta) = \beta(\theta_0) = \mathbb{P}_{\theta_0}(T > c).$$

By setting $\mathbb{P}_{\theta_0}(T > c) = \alpha$, we ensure that $\mathbb{P}_\theta(T > c) \leq \alpha$ for all $\theta < \theta_0$. Thus, c correctly guarantees that the size of the UMP test is α .

2 Misspecified Maximum Likelihood (50 points)

Let $\{X_i : 1, 2, \dots, n\}$ be a random sample from $X \sim \mathbb{P}_{\theta_0}$ with $\mathbb{P}_{\theta_0} \in \mathcal{P} = \{\mathbb{P}_\theta : \theta \in \Theta \subseteq \mathbb{R}\}$, where $F(x) = \mathbb{P}_\theta[X_i \leq x]$ is absolutely continuous with Lebesgue density $f(x; \theta)$. Let $f(x) = f(x; \theta_0)$ to save notation. Consider another family of Lebesgue densities, $\{g(x; \theta) : \theta \in \Theta \subseteq \mathbb{R}\}$, which may not be equal to $f(x; \theta)$. A *possibly misspecified* Maximum Likelihood Estimator (MLE) is

$$\check{\theta}_n = \arg \max_{\theta \in \Theta} \frac{1}{n} \sum_{i=1}^n \log g(X_i; \theta),$$

and the associated population optimization problem is

$$\check{\theta}_0 = \arg \max_{\theta \in \Theta} \mathbb{E}_{\theta_0}[\log g(X; \theta)] = \arg \max_{\theta \in \Theta} \int [\log g(x; \theta)] f(x) dx.$$

assuming the “usual” regularity conditions hold. You will be asked to state and impose additional assumptions (or regularity conditions) as needed below.

1. Providing appropriate regularity conditions, show that

$$\mathbb{E}_{\theta_0} \left[\log \left(\frac{g(X; \theta)}{f(X)} \right) \right] = \int \log \left(\frac{g(x; \theta)}{f(x)} \right) f(x) dx \leq 0,$$

with equality if and only if $g(x; \theta) = f(x)$.

Solution: Assuming we can interchange derivatives and expectations, by Jensen’s inequality and concavity of the $\log(\cdot)$ function,

$$\mathbb{E}_{\theta_0} \left[\log \left(\frac{g(X; \theta)}{f(X)} \right) \right] \leq \log \left(\int \frac{g(x; \theta)}{f(x)} f(x) dx \right) = \log \left(\int g(x; \theta) dx \right) = \log 1 = 0.$$

2. Use this result to justify the following interpretation for the pseudo-true value $\check{\theta}_0$:

$$\check{\theta}_0 = \arg \min_{\theta \in \Theta} D(g(\cdot, \theta), f), \quad D(g(\cdot, \theta), f) = \int [\log f(x)] f(x) dx - \int [\log g(x; \theta)] f(x) dx.$$

The “metric” $D(g(\cdot, \theta), f)$ is called the *Kullback-Leibler* (K-L) distance. Can you give an interpretation to the (possibly) misspecified MLE based on this criterion?

Solution: $D(g(\cdot, \theta), f)$ can be considered as a measure of fit between the two distributions, the closer this quantity is to zero the better the fit. The Kullback-Leibler criterion is commonly used for model selection. Note that $\check{\theta}_0 = \arg \min_{\theta \in \Theta} D(g(\cdot, \theta), f)$ implies that $g(\cdot, \check{\theta}_0)$ is the best fitting distribution using the K-L criterion. This does not mean it is the best fitting distribution according to another criterion. Indeed, if we used a different distance measure, we would likely obtain a different best fitting distribution. There are many different information criterion. The motivation for the KL criterion comes from the likelihood. However, in the

model misspecification setup there are alternative methods to likelihood methods, which can be used to find the best fitting distribution (these methods may be more robust, such as the Renyi information criterion).

3. Providing appropriate regularity conditions, show that

$$\mathbb{E}_{\theta_0} \left[\left. \frac{\partial}{\partial \theta} \log g(X; \theta) \right|_{\theta=\check{\theta}_0} \right] = 0.$$

What happens when $g(x; \check{\theta}_0) = f(x)$ or when $g(x; \theta) = f(x; \theta)$?

Solution: Since

$$\check{\theta}_0 = \arg \max_{\theta \in \Theta} \mathbb{E}_{\theta_0} [\log g(X; \theta)],$$

assuming differentiability,

$$0 = \frac{\partial}{\partial \theta} \mathbb{E}_{\theta_0} [\log g(X; \theta)]|_{\theta=\check{\theta}_0} = \mathbb{E}_{\theta_0} \left[\left. \frac{\partial}{\partial \theta} \log g(X; \theta) \right|_{\theta=\check{\theta}_0} \right]$$

where the second equality assumes interchangeability of derivatives and expectations. When $g(x; \check{\theta}_0) = f(x)$ or when $g(x; \theta) = f(x; \theta)$, we get $\check{\theta}_0 = \theta_0$.

4. Providing appropriate regularity conditions, show that

$$\sqrt{n}(\check{\theta}_n - \check{\theta}_0) = -\frac{1}{\sqrt{n}} \sum_{i=1}^n \psi(X_i) + o_{\mathbb{P}}(1) \rightarrow_d \mathcal{N}(0, V(\check{\theta}_0)),$$

where $\mathbb{E}_{\theta_0} [\psi(X_i)] = 0$ and $\mathbb{E}_{\theta_0} [\psi(X_i)^2] = V(\check{\theta}_0)$.

Give the exact form of the influence function $\psi(x)$.

Solution: Start by assuming that

$$\sup_{\theta \in \Theta} \left| \frac{1}{n} \sum_{i=1}^n \log g(X_i; \theta) - \mathbb{E}_{\theta_0} [\log g(X; \theta)] \right| = o_{\mathbb{P}}(1)$$

to ensure consistency of $\check{\theta}_n$ for $\check{\theta}_0$. Then, assuming differentiability,

$$\begin{aligned} 0 &= \frac{1}{n} \sum_{i=1}^n \frac{\partial}{\partial \theta} \log g(X_i; \check{\theta}_n) \\ &= \frac{1}{n} \sum_{i=1}^n \frac{\partial}{\partial \theta} \log g(X_i; \check{\theta}_0) + \frac{1}{n} \sum_{i=1}^n \frac{\partial^2}{\partial \theta^2} \log g(X_i; \check{\theta}_n) (\check{\theta}_n - \check{\theta}_0) \end{aligned}$$

where $\tilde{\theta}_n \rightarrow_{\mathbb{P}} \check{\theta}_0$ by consistency of $\check{\theta}_n$. Now, define

$$H_n(\theta) = \frac{1}{n} \sum_{i=1}^n \frac{\partial^2}{\partial \theta^2} \log g(X_i; \theta), \quad H(\theta) = \mathbb{E}_{\theta_0} \left[\frac{\partial^2}{\partial \theta^2} \log g(X_i; \theta) \right]$$

and suppose

$$\sup_{\theta \in \Theta} |H_n(\theta) - H(\theta)| = o_{\mathbb{P}}(1).$$

Then,

$$\begin{aligned} \left| H_n(\tilde{\theta}_n) - H(\check{\theta}_0) \right| &\leq \left| H_n(\tilde{\theta}_n) - H(\tilde{\theta}_n) \right| + \left| H(\tilde{\theta}_n) - H(\check{\theta}_0) \right| \\ &\leq \sup_{\theta \in \Theta} |H_n(\theta) - H(\theta)| + \left| H(\tilde{\theta}_n) - H(\check{\theta}_0) \right| \\ &= o_{\mathbb{P}}(1) + o_{\mathbb{P}}(1) = o_{\mathbb{P}}(1) \end{aligned}$$

where the first term goes to zero by assumption and the second one goes to zero by consistency of $\tilde{\theta}_n$. Therefore,

$$\begin{aligned} \sqrt{n}(\check{\theta}_n - \check{\theta}_0) &= \left[\frac{1}{n} \sum_{i=1}^n \frac{\partial}{\partial \theta} \log g(X_i; \tilde{\theta}_n) \right]^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^n \frac{\partial}{\partial \theta} \log g(X_i; \check{\theta}_0) \\ &= H_0^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^n \frac{\partial}{\partial \theta} \log g(X_i; \check{\theta}_0) + o_{\mathbb{P}}(1) \end{aligned}$$

by the CMT and using that the second sum is $O_{\mathbb{P}}(1)$ and where $H_0 = H(\check{\theta}_0)$. It follows that

$$\sqrt{n}(\check{\theta}_n - \check{\theta}_0) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \psi(X_i) + o_{\mathbb{P}}(1)$$

with

$$\psi(x) = H_0^{-1} \frac{\partial}{\partial \theta} \log g(x; \check{\theta}_0).$$

5. Show that $V(\check{\theta}_0) = H(\check{\theta}_0)^{-1} \Sigma(\check{\theta}_0) H(\check{\theta}_0)^{-1}$, and give the exact form of the variance components $H(\theta)$ and $\Sigma(\theta)$.

What happens when $g(x; \check{\theta}_0) = f(x)$ or when $g(x; \theta) = f(x; \theta)$?

Solution: Using the results from the previous part,

$$\mathbb{V}_{\theta_0}[\psi(X_i)] = H_0^{-1} \Sigma_0 H_0^{-1}, \quad \Sigma_0 = \Sigma(\check{\theta}_0) \quad \Sigma(\theta) = \mathbb{E}_{\theta_0} \left[\left(\frac{\partial}{\partial \theta} \log g(x; \theta) \right)^2 \right].$$

When $g(x; \check{\theta}_0) = f(x)$ or $g(x; \theta) = f(x; \theta)$, the sandwich variance reduces to the inverse of the information, that is, $V_0 = \mathcal{I}_0^{-1}$.

6. Propose a consistent estimator of the asymptotic variance $V(\check{\theta}_0)$.

Construct a feasible, asymptotically pivotal statistic.

What happens when $g(x; \check{\theta}_0) = f(x)$ or when $g(x; \theta) = f(x; \theta)$?

Solution: By consistency of $\check{\theta}_n$, we estimate the variance terms consistently using $\hat{H} = \hat{H}(\check{\theta}_n)$ and $\hat{\Sigma} = \hat{\Sigma}(\check{\theta}_n)$, where

$$\hat{H}(\theta) = \frac{1}{n} \sum_{i=1}^n \frac{\partial^2}{\partial \theta^2} \log g(X_i; \theta) \quad \text{and} \quad \hat{\Sigma}(\theta) = \frac{1}{n} \sum_{i=1}^n \left(\frac{\partial}{\partial \theta} \log g(x; \theta) \right)^2.$$

We can use two equivalent asymptotically pivotal statistics,

$$T(\check{\theta}_0) = \hat{V}^{-1/2} \sqrt{n}(\check{\theta}_n - \check{\theta}_0) \rightarrow_d \mathcal{N}(0, 1)$$

and

$$Q(\check{\theta}_0) = n(\check{\theta}_n - \check{\theta}_0)' \hat{V}^{-1} (\check{\theta}_n - \check{\theta}_0) \rightarrow_d \chi_1^2,$$

where $\hat{V} = \hat{H}^{-1} \hat{\Sigma} \hat{H}^{-1}$.

7. Suppose that $X_i = \mathbb{1}(\theta_0 - \epsilon_i \geq 0)$ with $\{\epsilon_i : 1, 2, \dots, n\}$ be a random sample from $\epsilon \sim \mathcal{N}(0, 1)$.

- (a) Show that $\mathcal{P} = \{\text{Bernoulli}(\Phi(\theta)) : \theta \in \Theta \subseteq \mathbb{R}\}$.

Find the (correctly specified) maximum likelihood estimator $\hat{\theta}_{\text{ML}}$ of θ_0 .

Find the asymptotic distribution of $\sqrt{n}(\hat{\theta}_{\text{ML}} - \theta_0)$, and give the exact form of the asymptotic variance component(s).

Solution: we have that $X_i \sim \text{Bernoulli}(p_0)$ where $p_0 = \Phi(\theta_0)$. The log-likelihood is given by

$$\ell_n(\theta) = \log \Phi(\theta) \sum_{i=1}^n X_i + \log(1 - \Phi(\theta)) \left(n - \sum_{i=1}^n X_i \right)$$

The FOC is

$$\frac{\bar{X}_n}{\Phi(\hat{\theta})} - \frac{(n - \bar{X}_n)}{1 - \Phi(\hat{\theta})} = 0 \quad \Rightarrow \quad \hat{\theta}_{\text{ML}} = \Phi^{-1}(\bar{X}_n)$$

By the delta method, given that $\mathbb{E}_{\theta_0}[X_i] = \Phi(\theta_0)$ and $\mathbb{V}_{\theta_0}[X_i] = \Phi(\theta_0)(1 - \Phi(\theta_0))$,

$$\sqrt{n}(\hat{\theta}_{\text{ML}} - \theta_0) \rightarrow_d \mathcal{N}(0, g'(\Phi(\theta_0))^2 \Phi(\theta_0)(1 - \Phi(\theta_0)))$$

where

$$g(z) = \Phi^{-1}(z), \quad g'(z) = \frac{1}{\phi(\Phi^{-1}(z))}, \quad g'(\Phi(\theta_0)) = \frac{1}{\phi(\theta_0)}$$

so the asymptotic variance is

$$V_0 = \frac{\Phi(\theta_0)(1 - \Phi(\theta_0))}{\phi(\theta_0)^2}$$

(b) Suppose a researcher employs the misspecified model $\epsilon \sim \text{Uniform}(0, 1)$.

Find the misspecified maximum likelihood estimator $\check{\theta}_n$ of $\check{\theta}_0$, and compute the form of pseudo-true value $\check{\theta}_0$ in this case.

Find the asymptotic distribution of $\sqrt{n}(\check{\theta}_n - \check{\theta}_0)$, and give the exact form of the asymptotic variance component(s).

Solution: under the misspecified model $g(x, \theta) = \theta^x(1 - \theta)^{1-x}$. The misspecified MLE is solves

$$\frac{\bar{X}_n}{\hat{\theta}} - \frac{1 - \bar{X}_n}{1 - \hat{\theta}} = 0 \quad \Rightarrow \quad \check{\theta}_n = \bar{X}_n$$

On the other hand, $\check{\theta}_0$ solves

$$\frac{\Phi(\theta_0)}{\check{\theta}_0} - \frac{1 - \Phi(\theta_0)}{1 - \check{\theta}_0} = 0 \quad \Rightarrow \quad \check{\theta}_0 = \Phi(\theta_0)$$

It follows by the CLT that

$$\sqrt{n}(\check{\theta}_n - \check{\theta}_0) \rightarrow_d \mathcal{N}(0, V), \quad V = \Phi(\theta_0)(1 - \Phi(\theta_0)).$$

8. Consider the hypothesis testing problem:

$$\mathbf{H}_0 : g(x; \check{\theta}_0) = f(x) \quad \text{vs.} \quad \mathbf{H}_1 : g(x; \check{\theta}_0) \neq f(x).$$

In words, $\mathbf{H}_0$ corresponds to correct specification of the likelihood function, and $\mathbf{H}_1$ corresponds to misspecification of the likelihood function.

(a) Under minimal regularity conditions,

$$-\mathbb{E}_{\theta_0} \left[\frac{\partial^2}{\partial \theta^2} \log g(X; \theta) \Big|_{\theta=\check{\theta}_0} \right] \neq \mathbb{E}_{\theta_0} \left[\left(\frac{\partial}{\partial \theta} \log g(X; \theta) \Big|_{\theta=\check{\theta}_0} \right)^2 \right]$$

unless $g(x; \check{\theta}_0) = f(x)$. Using this result, propose an infeasible hypothesis test for misspecification of the likelihood function.

Solution: This is known as the **Information Matrix Test**. The intuition is as follows: if the likelihood is correctly specified, then the information matrix identity holds, but if the likelihood model is misspecified then the expectation of the Hessian is not equal to the expectation of the outer product of the score. To formalize the idea, define

$$D(\theta) = \mathbb{E}[d(X_i, \theta)], \quad d(X_i, \theta) = -\frac{\partial^2}{\partial \theta^2} \log g(X; \theta) - \left(\frac{\partial}{\partial \theta} \log g(X; \theta) \right)^2$$

Clearly, $D(\check{\theta}_0) = 0$ under $\mathbf{H}_0$, and $D(\check{\theta}_0) \neq 0$ under $\mathbf{H}_1$. Thus, a sample analogue of $|D(\check{\theta}_0)|$ should be “small” under $\mathbf{H}_0$, and “large” under $\mathbf{H}_1$. Therefore, the infeasible

testing problem is: Reject H_0 if (and only if) $|D(\check{\theta}_0)| > 0$.

- (b) Propose a feasible testing procedure for misspecification of the likelihood function.

Solution: A sample analogue of $D(\theta)$ is $\bar{D}_n(\theta) = \frac{1}{n} \sum_{i=1}^n d(X_i, \theta)$, which is a sample average of i.i.d. random variables. Therefore, under H_0 , we have

$$T_n(\check{\theta}_n) = \frac{\bar{D}_n(\check{\theta}_n)}{\sqrt{S_n(\check{\theta}_n)/n}} \rightsquigarrow \text{Normal}(0, 1), \quad S_n^2(\check{\theta}_n) = \frac{1}{n-1} \sum_{i=1}^n (d(X_i, \check{\theta}_n) - \bar{D}_n)^2,$$

provided standard regularity conditions hold (e.g., $\mathbb{E}_{\theta_0}[|d(X_i, \theta)|^2] < \infty$ uniformly in θ , differentiability and integrability of $\theta \mapsto d(X_i, \theta)$ uniformly in θ , etc.). The feasible α level hypothesis test rejects H_0 if (and only if) $T_n(\check{\theta}_n) > \Phi(1 - \alpha/2)$, where $\Phi(u) = \mathbb{P}[\text{Normal}(0, 1) \leq u]$.

Table 1.1. Discrete Distributions on $\mathcal{R}$

Uniform	p.d.f.	$1/m, x = a_1, \dots, a_m$
	m.g.f.	$\sum_{j=1}^m e^{a_j t}/m, t \in \mathcal{R}$
$DU(a_1, \dots, a_m)$	Expectation	$\sum_{j=1}^m a_j/m$
	Variance	$\sum_{j=1}^m (a_j - \bar{a})^2/m, \bar{a} = \sum_{j=1}^m a_j/m$
	Parameter	$a_i \in \mathcal{R}, m = 1, 2, \dots$
Binomial	p.d.f.	$\binom{n}{x} p^x (1-p)^{n-x}, x = 0, 1, \dots, n$
	m.g.f.	$(pe^t + 1 - p)^n, t \in \mathcal{R}$
$Bi(p, n)$	Expectation	np
	Variance	$np(1-p)$
	Parameter	$p \in [0, 1], n = 1, 2, \dots$
Poisson	p.d.f.	$\theta^x e^{-\theta}/x!, x = 0, 1, 2, \dots$
	m.g.f.	$e^{\theta(e^t - 1)}, t \in \mathcal{R}$
$P(\theta)$	Expectation	θ
	Variance	θ
	Parameter	$\theta > 0$
Geometric	p.d.f.	$(1-p)^{x-1}p, x = 1, 2, \dots$
	m.g.f.	$pe^t/[1 - (1-p)e^t], t < -\log(1-p)$
$G(p)$	Expectation	$1/p$
	Variance	$(1-p)/p^2$
	Parameter	$p \in [0, 1]$
Hyper-geometric	p.d.f.	$\binom{n}{x} \binom{m}{r-x} / \binom{N}{r}$
		$x = 0, 1, \dots, \min\{r, n\}, r - x \leq m$
$HG(r, n, m)$	m.g.f.	No explicit form
	Expectation	rn/N
	Variance	$rn m(N-r)/[N^2(N-1)]$
	Parameter	$r, n, m = 1, 2, \dots, N = n + m$
Negative binomial	p.d.f.	$\binom{x-1}{r-1} p^r (1-p)^{x-r}, x = r, r+1, \dots$
	m.g.f.	$p^r e^{rt}/[1 - (1-p)e^t]^r, t < -\log(1-p)$
$NB(p, r)$	Expectation	r/p
	Variance	$r(1-p)/p^2$
	Parameter	$p \in [0, 1], r = 1, 2, \dots$
Log-distribution	p.d.f.	$-(\log p)^{-1} x^{-1} (1-p)^x, x = 1, 2, \dots$
	m.g.f.	$\log[1 - (1-p)e^t]/\log p, t \in \mathcal{R}$
$L(p)$	Expectation	$-(1-p)/(p \log p)$
	Variance	$-(1-p)[1 + (1-p)/\log p]/(p^2 \log p)$
	Parameter	$p \in (0, 1)$

All p.d.f.'s are w.r.t. counting measure.

Table 1.2. Distributions on $\mathcal{R}$ with Lebesgue p.d.f.'s

Uniform	p.d.f.	$(b-a)^{-1}I_{(a,b)}(x)$
	m.g.f.	$(e^{bt} - e^{at})/[(b-a)t], \quad t \in \mathcal{R}$
$U(a, b)$	Expectation	$(a+b)/2$
	Variance	$(b-a)^2/12$
	Parameter	$a, b \in \mathcal{R}, \quad a < b$
Normal	p.d.f.	$\frac{1}{\sqrt{2\pi}\sigma}e^{-(x-\mu)^2/2\sigma^2}$
	m.g.f.	$e^{\mu t + \sigma^2 t^2/2}, \quad t \in \mathcal{R}$
$N(\mu, \sigma^2)$	Expectation	μ
	Variance	σ^2
	Parameter	$\mu \in \mathcal{R}, \quad \sigma > 0$
Exponential	p.d.f.	$\theta^{-1}e^{-(x-a)/\theta}I_{(a,\infty)}(x)$
	m.g.f.	$e^{at}(1-\theta t)^{-1}, \quad t < \theta^{-1}$
$E(a, \theta)$	Expectation	$\theta + a$
	Variance	θ^2
	Parameter	$\theta > 0, \quad a \in \mathcal{R}$
Chi-square	p.d.f.	$\frac{1}{\Gamma(k/2)2^{k/2}}x^{k/2-1}e^{-x/2}I_{(0,\infty)}(x)$
	m.g.f.	$(1-2t)^{-k/2}, \quad t < 1/2$
χ_k^2	Expectation	k
	Variance	$2k$
	Parameter	$k = 1, 2, \dots$
Gamma	p.d.f.	$\frac{1}{\Gamma(\alpha)\gamma^\alpha}x^{\alpha-1}e^{-x/\gamma}I_{(0,\infty)}(x)$
	m.g.f.	$(1-\gamma t)^{-\alpha}, \quad t < \gamma^{-1}$
$\Gamma(\alpha, \gamma)$	Expectation	$\alpha\gamma$
	Variance	$\alpha\gamma^2$
	Parameter	$\gamma > 0, \quad \alpha > 0$
Beta	p.d.f.	$\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}x^{\alpha-1}(1-x)^{\beta-1}I_{(0,1)}(x)$
	m.g.f.	No explicit form
$B(\alpha, \beta)$	Expectation	$\alpha/(\alpha+\beta)$
	Variance	$\alpha\beta/[(\alpha+\beta+1)(\alpha+\beta)^2]$
	Parameter	$\alpha > 0, \quad \beta > 0$
Cauchy	p.d.f.	$\frac{1}{\pi\sigma}\left[1+\left(\frac{x-\mu}{\sigma}\right)^2\right]^{-1}$
	ch.f.	$e^{\sqrt{-1}\mu t - \sigma t }$
$C(\mu, \sigma)$	Expectation	Does not exist
	Variance	Does not exist
	Parameter	$\mu \in \mathcal{R}, \quad \sigma > 0$

Table 1.2. (continued)

t-distribution	p.d.f.	$\frac{\Gamma[(n+1)/2]}{\sqrt{n\pi}\Gamma(n/2)} \left(1 + \frac{x^2}{n}\right)^{-(n+1)/2}$
	ch.f.	No explicit form
t_n	Expectation	0, $(n > 1)$
	Variance	$n/(n-2)$, $(n > 2)$
	Parameter	$n = 1, 2, \dots$
F-distribution	p.d.f.	$\frac{n^{n/2}m^{m/2}\Gamma[(n+m)/2]x^{n/2-1}}{\Gamma(n/2)\Gamma(m/2)(m+nx)^{(n+m)/2}}I_{(0,\infty)}(x)$
	ch.f.	No explicit form
$F_{n,m}$	Expectation	$m/(m-2)$, $(m > 2)$
	Variance	$2m^2(n+m-2)/[n(m-2)^2(m-4)]$, $(m > 4)$
	Parameter	$n = 1, 2, \dots$, $m = 1, 2, \dots$
Log-normal	p.d.f.	$\frac{1}{\sqrt{2\pi}\sigma}x^{-1}e^{-(\log x - \mu)^2/2\sigma^2}I_{(0,\infty)}(x)$
	ch.f.	No explicit form
$LN(\mu, \sigma^2)$	Expectation	$e^{\mu+\sigma^2/2}$
	Variance	$e^{2\mu+\sigma^2}(e^{\sigma^2} - 1)$
	Parameter	$\mu \in \mathcal{R}$, $\sigma > 0$
Weibull	p.d.f.	$\frac{\alpha}{\theta}x^{\alpha-1}e^{-x^\alpha/\theta}I_{(0,\infty)}(x)$
	ch.f.	No explicit form
$W(\alpha, \theta)$	Expectation	$\theta^{1/\alpha}\Gamma(\alpha^{-1} + 1)$
	Variance	$\theta^{2/\alpha} \left\{ \Gamma(2\alpha^{-1} + 1) - [\Gamma(\alpha^{-1} + 1)]^2 \right\}$
	Parameter	$\theta > 0$, $\alpha > 0$
Double Exponential	p.d.f.	$\frac{1}{2\theta}e^{- x-\mu /\theta}$
	m.g.f.	$e^{\mu t}/(1 - \theta^2 t^2)$, $ t < \theta^{-1}$
$DE(\mu, \theta)$	Expectation	μ
	Variance	$2\theta^2$
	Parameter	$\mu \in \mathcal{R}$, $\theta > 0$
Pareto	p.d.f.	$\theta a^\theta x^{-(\theta+1)}I_{(a,\infty)}(x)$
	ch.f.	No explicit form
$Pa(a, \theta)$	Expectation	$\theta a/(\theta - 1)$, $(\theta > 1)$
	Variance	$\theta a^2/[(\theta - 1)^2(\theta - 2)]$, $(\theta > 2)$
	Parameter	$\theta > 0$, $a > 0$
Logistic	p.d.f.	$\sigma^{-1}e^{-(x-\mu)/\sigma}/[1 + e^{-(x-\mu)/\sigma}]^2$
	m.g.f.	$e^{\mu t}\Gamma(1 + \sigma t)\Gamma(1 - \sigma t)$, $ t < \sigma^{-1}$
$LG(\mu, \sigma)$	Expectation	μ
	Variance	$\sigma^2\pi^2/3$
	Parameter	$\mu \in \mathcal{R}$, $\sigma > 0$